

Graphene Aluminum Composites for Use in Radiators for Nuclear Electric Propulsion

Principal Investigator

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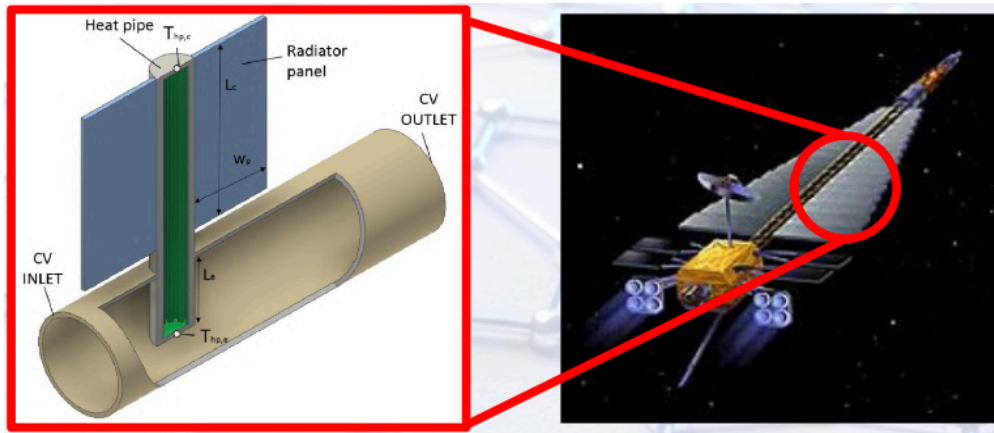
Subject Matter Expert

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Core Team Members

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Jordan Ando, Candace Do, Harsha Antony,
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2015 NASA Technology Roadmap Identifier TX 12.1.4



Abstract

One of the biggest challenges when it comes to traveling with low gravity is propulsion. Factors including energy efficiency and weight are important aspects to consider before any spacecraft is launched. Currently, nuclear electric propulsion systems use carbon as the material for the radiators. While this is the state of the art technology currently being implemented, we propose replacing the carbon fins with an aluminum-graphene composite to decrease the weight, launch cost, and fuel required to send one of these propulsion systems into space. The properties of an aluminum-graphene composite make it a more suitable choice than current solutions such as carbon radiator fins. Aluminum-graphene composites are significantly lighter than the carbon material currently used, and they are easier to manufacture since the technology needed to manufacture carbon for radiators has not been fully developed. This enhanced material also provides great heat conductivity as opposed to other composites and materials. This can lead to higher levels of fuel efficiency for its lightweight properties. This material is also cost-effective, creating massive overall benefits towards nuclear propulsion. Researching aluminum-graphene composites and delving into their capabilities can help push the next wave of technology for nuclear electric propulsion systems.

Technology Merit and Work

Current State of the Art

<https://ntrs.nasa.gov/citations/20130001608>

For high-power nuclear-electric spacecraft, the radiator can account for 40% or more of the power system mass and a large fraction of the total vehicle mass. Improvements in the heat rejection per unit mass rely on lower-density and higher-thermal conductivity materials. Current radiators achieve near-ideal surface radiation through high-emissivity coatings, so improvements in heat rejection per unit area can be accomplished only by raising the temperature at which heat is rejected. We have been investigating materials that have the potential to deliver significant reductions in mass density and significant improvements in thermal conductivity, while expanding the feasible range of temperature for heat rejection up to 1000 K and higher. The presentation will discuss the experimental results and models of the heat transfer in matrix-free carbon fiber fins. Thermal testing of other carbon-based fin materials including carbon nanotube cloth and a carbon nanotube composite will also be presented.

Thermal control is important to all spacecraft. One vital component of many thermal control systems is a radiator. Radiators are typically manufactured out of a thermally conductive material and use coolant loops to move excess heat out of the spacecraft. Radiators should contain a material that has high thermal conductivity. This will allow for heat transfer to the environment, and a high specific heat, which reduces the radiator's temperature. Current space radiators, such as those present on the International Space Station (ISS), are made of aluminum. The ISS has three radiators with dimensions measuring 10.24 x 44.62 meters and a mass of 740.7 kg.

There is a clear need for lighter and more efficient radiators. Reducing mass can drastically reduce launch costs and fuel requirements for future missions, while more thermally efficient radiators widen potential applications. For instance, nuclear electric propulsion systems, which are currently under development, will require high thermal conduction to dissipate the massive amounts of generated heat. According to Hyers et. al (2013), it is not unreasonable for a radiator to account for over 40% of the mass of a nuclear-electric propulsion system. This means that reducing radiator mass is a vital field of research for future spacecraft.

The current state of the art method being researched for similar applications is fully carbon radiator fins. While carbon has some advantages as a material, it is known for being incredibly thermally conductive, lightweight, and has high tensile strength. It is expensive and has yet to be used to make radiators in large-scale production. Manufacturing with carbon is difficult since it cannot be machined traditionally, so new methods must be developed alongside the research of the material.

Description of Technology

This technology concept involves taking aluminum/graphene (Al/G) composites and applying them to radiators. Several research groups have looked into various Al/G composites and demonstrated their enhanced mechanical and thermal properties. Thus, these composites are likely well-suited for applications in radiators and in other areas that require thermally conductive, lightweight materials. The small amount of graphene required for an effective composite makes it a preferable alternative to fully carbon solutions.

Al/G composites are well poised to solve pain points regarding mass and thermal capabilities. Graphene, a lightweight carbon allotrope, has been long known to have promising thermal properties. Recently, graphene, composited with aluminum at a mere 0.3% wt, has been demonstrated to increase aluminum’s thermal conductivity by 15.4% to 280 W/mK, and specific heat by 9.1% (Zhang et. al, 2018). Increases in compressive strength of 25.6% were observed over pure aluminum manufactured in the same way. Moreover, at a density of 1.124 g/cm³, Al/G composites are 40% lighter per unit volume than the current state of the art aluminum alloy 2219-T6.

An Al/G composite will be created to replace aluminum composites currently used in radiators for nuclear electric propulsion. The Al/G composite will aim to reduce the areal density of radiators to the desired 2-4 kg/m², while also having high thermal conductivity and specific heat. Specific key performance parameters are presented in Table 1.

While it is true that a full carbon fin would exceed the heat transfer capabilities of an Al/G composite fin, current carbon fiber tested for radiator applications by Tombulian et al. has a density of 2.21 g/cm³, suggesting that Al/G, with its lower density, should also be able to reach the desired areal density of 2-4 kg/m² (Tombulian et al., 2014) and potentially surpass the low-mass benefits of carbon prototypes. Moreover, the emissivity of carbon, an important factor in calculating radiated heat using the Stefan-Boltzmann law, is generally accepted as low, leaving room for improvement with other materials. Unlike fully carbon options, Zhang et al. note that their Al/G composite is fully machinable, making manufacturing radiators or other parts a much more straightforward, familiar process.

Key performance parameters are outlined below in Table 1.

Table 1: Key Performance Parameters

Key Performance Parameter	Metric
Density	1.124 g/cm ³
Compressive strength	260 Mpa
Thermal conductivity	280 W/mK
Specific heat	0.987 J/g/K

Challenges and Risks

One potential challenge of manufacturing an aluminum graphene composite radiator is synthesizing the Al/G composite in large quantities. According to Dixit et al., multiple methods have previously been used for manufacturing. In chemical vapor deposition (CVD), intermittent graphene layers are placed on metal substrates, creating a graphene-metal composite. However, the powder-based processing of graphene such as this can result in a weak interface between the graphene and the metal. Another method is to disperse graphene into an aluminum matrix through ball-milling and reinforce with hot compaction. Due to the low density of graphene, however, graphene will float on top of the molten metal matrix. Another challenge is that unless the graphene is added to the metal matrix in a strong vacuum, the graphene can become oxidized to form graphene oxide, which is a toxic compound.

The method most commonly used in manufacturing aluminum graphene composites is friction stir processing (FSP). FSP is used to “produce very high shear strains (around 40) in the matrix that ultimately exfoliates the embedded graphite particles leading to a rapidly formed assembly of the graphene layers surrounded by aluminum crystallite surfaces without allowing any oxidation or metal-carbide type bonds to form” (Dixit et al.). Ultimately, the challenges in manufacturing may best be approached by outsourcing the manufacturing of the composite material and focusing available lab facilities on testing the material. This way, the primary risk comes is that the manufacturing company is unsuccessful, and it also allows for more time and resources to be devoted to designing the radiator.

Obstacles also exist for the development of radiators. Al/G composites, while having several desirable properties for radiators for nuclear-electric propulsion, come with a few unknowns. For a material to be tenable for radiator construction, it needs to be able to withstand temperatures exceeding 1000 K (Hyers et al., 2013). At this point, the high-temperature behavior of Al/G composites has yet to be fully investigated. It is noteworthy that Zhang et al. synthesized the material at ~873 K, and tested its specific heat through about ~1000 K. This shows that the material can hold up at this lower bound for nuclear-electric propulsion engines, but fails to examine higher temperatures or how such a radiator would fare over time. Thus, extensive testing of a radiator fin under extended thermal load will be an early required step before further development.

Technology Merit

The creation and application of Al/G composites are currently at TRL 2. Various research institutions have, over the last decade, managed to manufacture composites with basic desirable qualities. The phase wherein a radiator could be manufactured has come, provided a research group can obtain or manufacture Al/G composites on a large enough scale. Testing a radiator prototype serves a dual purpose, as it demonstrates the material’s ability to function as a radiator, while also allowing testing of the material’s emissivity, durability under a thermal load, and other properties important to peak radiator performance. Such tests would act as a foil to other NASA efforts to develop viable nuclear-electric propulsion systems and their radiators, as well as provide proof that these newly developed composites have suitable applications.

Alignment to Agency and/or National Needs

This concept will not only lower mission costs, but increase efficiency in thermal conductivity by mass, specific heat, compressive strength, and weight of manufactured radiators. By applying the concept,

NASA will be able to further fund their projects in other crucial aspects while increasing the effectiveness of their radiators. This may result in an increase in space exploration due to the reliability and cost-efficiency of the Al/G composite radiators. This strategy of incorporating graphene to aluminum composites could be applied to many components of National requirements. This approach would be considered a “green” technology since it would lower the fuel requirements due to the lighter mass of its material. Along with this, Al/G composites could be employed by commercial spaceflight nationwide.

Work Plan

Due to time, cost, and space availability constraints, especially given that many labs have limited access due to COVID, research into this material is best started by focusing on two aspects: simulation of radiators and acquisition of Al/G composites, or the materials required for manufacturing. This process will occur in Fusion 360 for modeling, and COMSOL Multiphysics to run simulations. COMSOL is chosen due to its ability to combine multiple physics simulation tools, and to closely parameterize models to match anticipated material properties and conditions. This will allow for a prototype, small-scale design to be preliminarily tested and demonstrated functional before spending copious amounts of money on expensive composites. A digital testing environment also facilitates online collaboration, which is beneficial due to the distance between most team members. It also minimizes physical risk, and risk of damaging an expensive prototype. Finally, simulation of a radiator provides a sense as to whether or not a physical prototype is at all viable in the first place.

Once facilities are available, testing on a physical prototype can commence. Designs finalized in the simulation process will be machined at a team member’s home institution and tested under increasingly higher thermal loads. This will allow for the verification that a radiator made of Al/G composites functions at lower temperatures, and then at temperatures required for a nuclear-electric propulsion system. Finding lab facilities may pose a challenge due to limitations imposed by COVID, but the intent is for a majority of design and simulation to be done by the point that it may become possible.

During the entirety of this process, we would also be collaborating with our SME, who is a thermal expert, to help us and teach us during the development process. This will also allow us to make sure that all the steps we take fit NASA guidelines and meet professional standards which will be important in attempting further funding later down the line.

Project Management Approach

Management Approach

To approach the task, we will use a pseudo-waterfall approach to make sure that we are allocating resources while being as focused as possible. To do this, we have set deadlines and goals that will need to be completed before going on to the next task (as outlined in the project schedule and milestones below). One major factor in this approach ensures that every step of development is up to standards and expectations before moving on. For example, part of the process is designing a small scale prototype and then testing it. Without making sure that the small scale prototype is meeting, or can meet our expectations, we cannot move on. Another major benefit of the waterfall approach is our own schooling. Everyone within our team will be taking classes this fall, which will limit our workload. By prioritizing one part of the project at a time, we can allocate our time more efficiently, allowing for many team members to be working at the same time. Furthermore, due to school and other commitments, most team members will have set schedules, allowing us to set up regular meeting times. Finally, the most important

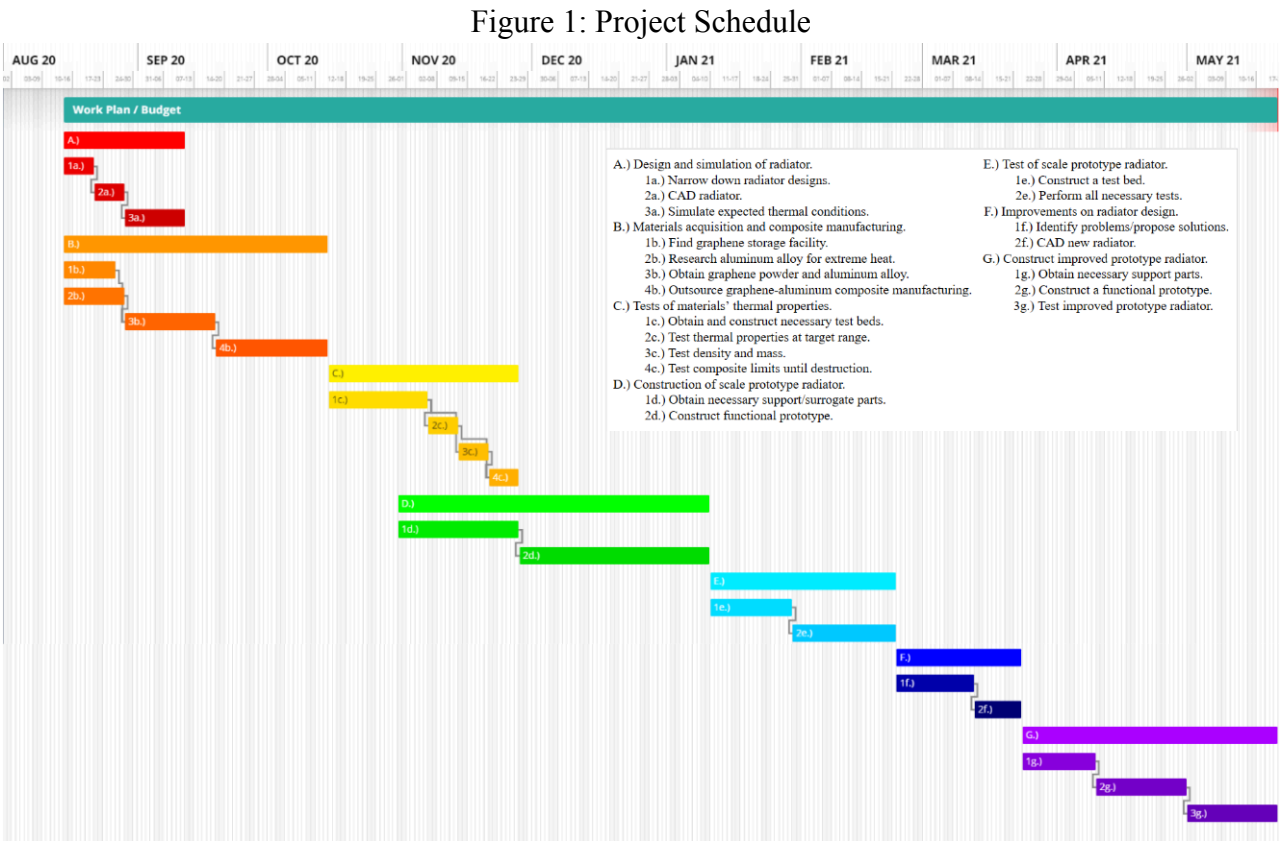
aspect of collaboration is proper communication. To do this we will use email, Discord, and Zoom to communicate and collaborate when we are unable to work in person. We have been using these methods to work together, and they have proven to be effective thus far. In implementing a solid and more consistent meeting schedule, we will be able to continue working well collaboratively.

Milestones

- 1. Development of aluminum/graphene composite.
- 2. Testing of aluminum/graphene composite for key performance parameters.
- 3. Design and simulation of radiator.
- 4. Development of radiator.
- 5. Testing of radiator for key performance parameters.

Project Schedule

A project schedule is outlined below in Figure 1.



Resources

Resources required for this project and their cost estimates are provided below in Table 2. The total cost estimate is \$7,468.50, less than the \$10,000 awarded.

Table 2: Resources and Cost Estimates

Resource	Cost Estimate
Aluminum/graphite composite	
Graphite powder	\$18.50 for 24 g (approx. \$0.77/g)

Aluminum sheets	\$450.00 for 8000 g (approx. \$0.056/g)
Outsourcing of aluminum/graphene composite manufacturing	\$4,000
Radiator prototype	
Liquid heat transfer system	\$2,500
Machining equipment	\$500 (rent from institution)
Software	
MATLAB	\$0 (institutional software)
COMSOL Multiphysics	\$0 (institutional software)
Autodesk Fusion 360	\$0 (free to install)
Total	\$7,468.50

Teaming and Workforce Development

Table 3 describes each core team member, their role on the project, their applicable experience, and the approximate time to be spent on the project.

Table 3: Core Team Members

Team member	Role	Applicable experience	Approximate time
Jordan Levin	Principal Investigator	CAD, Data Analysis, Materials Engineer Major	1-2 hours/day
Andrew Mills	Co-Principal Investigator	CAD, Data Analysis, Gantt Charts, Leadership	1-4 hours/day
Shannon Comstock	Co-Principal Investigator	Data Analysis, CAD	1-2 hours/day
Jordan Ando	Co-Project Manager	Physics simulation, coding, CAD	2-3 hours/day
Candace Do	Co-Project Manager	Coding, CAD	1-2 hours/day
Harsha Antony	Researcher	Coding, CAD	2-3 hours/day
Ariel Ben-Avides	Researcher	Data Analysis, CAD	1-2 hours/day
Khalid Mihlar	Researcher	Coding, CAD	1-2 hours/day
Justin Wong Lara	Researcher	Data Analysis, CAD	1-2 hours/day
Abigail Fellows	Project Management	Leadership, Team Building, Writing	1-2 hours/day

Further Potential

Once matured, our next step would be to pursue further funding, most likely through NASA. Through this funding, we would be able to continue development from a working prototype to a full scale system and eventually practical implementation in future NASA missions. Additionally, if we were to work more directly with NASA later down the line with the increased access to field experts, we would not only be able to develop the proposed radiator improvements but also potentially continue improving upon the proposed technology.

Appendix

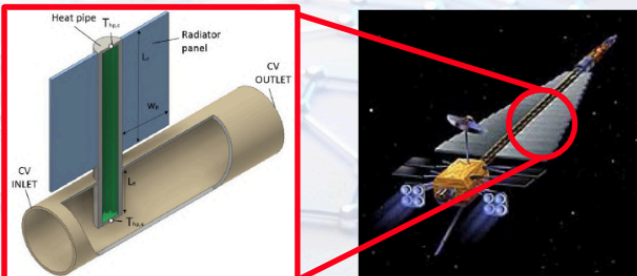
Quad Chart

Graphene Aluminum composites for use in Radiators for Nuclear Electric propulsion (GARNE)

PI: Jordan Levin, Team 18

Goal: Utilize a Graphene Aluminum composite in radiators to reduce materials required, improve thermal conductivity, strength, and specific heat capacity over current composites.

- Replace current aluminum composite used in radiators for heat dissipation with aluminum graphene composite
- Demonstrate that it improves upon efficiency compared to current composites
- Show that it can decrease mass/costs of radiators
- Create a prototype radiator applying Graphene Aluminum composites and test it.



L'SPACE NPWEE TEAM 18



Harsha Antony
Proposal Writer
Mechanical Engineering



Justin Wong
Lara
Proposal Editor
Civil Engineering



Khalid Mithar
Proposal Editor
Computer Engineering



Abigail Fellows
Project Manager
Political Science/Peace and Ethnic Studies



Candace Do
Co-Project Manager
Math & Labs Engineering



Andrew Mills
Co-Principal Investigator
Aerospace Engineering



Shannon Comstock
Co-Principal Investigator
Astrophysics



Jordan Levin
Principal Investigator
Materials Engineering



Adiel Ben-Avides
Proposal Editor
Engineering



Jordan Ando
Co-Project Manager
Astrophysics Engineering

Aluminum Graphene 0.3% wt Compared to Aluminum Alloy 2219-T6			
Al/G Thermal Conductivity	Al/G Specific Heat	Al/G Compressive Strength	Al/G Weight
Increase of 15.4%	Increase of 9.1%	Increase of 25.6%	40% Lighter

Key Performance Parameter	Metric
Density	1.124 g/cm ³
Compressive strength	260 Mpa
Thermal conductivity	280 W/mK
Specific heat	0.987 J/g/K

- Data shows that even with a small amount of Graphene, the composite improves on the state of the art.
- Due to the improvements in density and thermal conductivity, the same level of heat capacity can be reached with much lighter material. This could potentially significantly reduce cost.
- The final product still maintains both perfect machinability and mechanical properties allowing for it to be easily implemented.

New Technology Report (NTR)

	National Aeronautics and Space Administration	Disclosure of Invention and New Technology (Including Software)	Form Approved O.M.B. NO. 2700-0009	DATE
This is an important legal document. Carefully complete and forward to the Patent Representative (NASA in-house innovation) or New Technology Representative (contractor/grantee innovation) at NASA. Use of this report form by contractor/grantee is optional; however, an alternative format must at a minimum contain the information required herein. NASA in-house disclosures should be read, understood and signed by a technically competent witness in the witness signature block at the end of this form. In completing each section, use whatever detail deemed appropriate for a "full and complete disclosure." Contractors/Grantees please refer to the New Technology or Patent Rights – Retention by the Contractor clauses. When necessary, attach additional documentation to provide a full, detailed description.			CONTRACTOR CASE NO.	
		NASA CASE NO. (OFFICIAL USE ONLY)		
1. DESCRIPTIVE TITLE Graphene Aluminum composites for use in Radiators for Nuclear Electric propulsion (GARNE)				
2. INNOVATOR(S) <i>(For each innovator provide: Name, Title, Work Address, Work Phone Number, and Work E-mail Address. If multiple innovators, number each to match Box 5.)</i>				
1.) Jordan Levin, Principal Investigator, 1150 East University Drive, Tempe, AZ 85281, (847) 849-9214, jordan_levin@elcamino.edu 2.) Andrew Mills, Co-Principal Investigator, 1150 East University Drive, Tempe, AZ 85281, (310) 947-1346, andrew.mills@student.csulb.edu 3.) Shannon Comstock, Co-Principal Investigator, 1150 East University Drive, Tempe, AZ 85281, (818) 212-5736, Sac546@nau.edu 4.) Candace Do, Co-Project Manager, 1150 East University Drive, Tempe, AZ 85281, (425) 586-0537, candace.do@gmail.com				

3. INNOVATOR'S EMPLOYER WHEN INNOVATION WAS MADE <i>(For each innovator provide: Name, Division and Address of Employer, Organizational Code/Mail Code, and Contract/Grant Number if applicable. If multiple innovators, number each to match Box 5.)</i> NPWEE class with Arizona State University, 1150 East University Drive, Tempe, AZ 85281, LSPACE@asu.edu																									
4. PLACE OF PERFORMANCE <i>(Address(es) where innovation made)</i> NPWEE class with Arizona State University, 1150 East University Drive, Tempe, AZ 85281																									
5. EMPLOYER STATUS <i>(choose one for each innovator)</i> <table style="width: 100%; border: none;"> <tr> <td style="border-bottom: 1px solid black; width: 50%;"><u>Innovator #1</u></td> <td style="border-bottom: 1px solid black; width: 50%;"><u>Innovator #2</u></td> </tr> <tr> <td style="text-align: center;">CU</td> <td style="text-align: center;">CU</td> </tr> <tr> <td style="border-bottom: 1px solid black;"><u>Innovator #3</u></td> <td style="border-bottom: 1px solid black;"><u>Innovator #4</u></td> </tr> <tr> <td style="text-align: center;">CU</td> <td style="text-align: center;">CU</td> </tr> </table> GE = Government CU = College or University NP = Non-Profit Organization SB = Small Business Firm LE = Large Entity	<u>Innovator #1</u>	<u>Innovator #2</u>	CU	CU	<u>Innovator #3</u>	<u>Innovator #4</u>	CU	CU	6. ORIGIN <i>(Check all that apply and provide all applicable numbers. If multiple Contracts/Grants, etc., list Contract/Grant Numbers in Box 3 with applicable employer information.)</i> <table style="width: 100%; border: none;"> <tr> <td style="width: 60%;">☐ NASA In-house Org. Mail Code</td> <td style="width: 40%; text-align: right;">: WBS</td> </tr> <tr> <td>☒ Grant/Cooperative Agreement No. 80NSSC19M0186</td> <td style="text-align: right;">: WBS</td> </tr> <tr> <td>☐ Prime Contract No.</td> <td style="text-align: right;">: WBS</td> </tr> <tr> <td>Task No. Report No.</td> <td></td> </tr> <tr> <td>☐ Subcontractor; Subcontract Tier</td> <td style="text-align: right;">: WBS</td> </tr> <tr> <td>☐ Joint Effort (contractor, subcontractor and/or grantee contribution(s), and NASA in-house contribution)</td> <td style="text-align: right;">:</td> </tr> <tr> <td>☐ Multiple Effort (multiple contractor, subcontractor and/or grantee contributions, no NASA in-house contribution)</td> <td style="text-align: right;">:</td> </tr> <tr> <td>☐ Other (e.g., Space Act Agreement, MOA) No.</td> <td style="text-align: right;">: WBS</td> </tr> </table>	☐ NASA In-house Org. Mail Code	: WBS	☒ Grant/Cooperative Agreement No. 80NSSC19M0186	: WBS	☐ Prime Contract No.	: WBS	Task No. Report No.		☐ Subcontractor; Subcontract Tier	: WBS	☐ Joint Effort (contractor, subcontractor and/or grantee contribution(s), and NASA in-house contribution)	:	☐ Multiple Effort (multiple contractor, subcontractor and/or grantee contributions, no NASA in-house contribution)	:	☐ Other (e.g., Space Act Agreement, MOA) No.	: WBS
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☐ Other (e.g., Space Act Agreement, MOA) No.	: WBS																								

7. NASA CONTRACTING OFFICER'S TECHNICAL REPRESENTATIVE (COTR) John Dankanich	8. CONTRACTOR/GRANTEE NEW TECHNOLOGY REPRESENTATIVE (POC)
9. BRIEF ABSTRACT <i>(A general description of the innovation which describes its capabilities, but does not reveal details that would enable duplication or imitation of the innovation.)</i> Replacement material for state-of-the-art Nuclear Electric Propulsion Radiators. This material will be a composite of up and coming materials with tried and true aluminum alloys, designed to have excellent thermal properties and low mass.	

SECTION I – DESCRIPTION OF THE PROBLEM OR OBJECTIVE THAT MOTIVATED THE INNOVATION'S DEVELOPMENT *(Enter as appropriate: A. – General description of problem/objective; B. – Key or unique problem characteristics; C. – Prior art, i.e., prior techniques, methods, materials, or devices performing function of the innovation, or previous means for performing function of software; and D. – Disadvantages or limitation of prior art.)*

Usable nuclear electric propulsion requires light, high-efficiency radiators to function. The current state of the art materials meet the required thermal properties, but at a higher mass than is usable. Currently proposed radiators are based on molybdenum and since it improves per unit mass, too much mass is required to radiate heat sufficiently.

SECTION II – TECHNICALLY COMPLETE AND EASILY UNDERSTANDABLE DESCRIPTION OF INNOVATION DEVELOPED TO SOLVE THE PROBLEM OR MEET THE OBJECTIVE *(Enter as appropriate; existing reports, if available, may form a part of the disclosure, and reference thereto can be made to complete this description: A. – Purpose and description of innovation/software; B. – Identification of component parts or steps, and explanation of mode of operation of innovation/software preferably referring to drawings, sketches, photographs, graphs, flow charts, and/or parts or ingredient lists illustrating the components; C. – Functional operation; D. – Alternate embodiments of the innovation/software; E. – Supportive theory; F. – Engineering specifications; G. – Peripheral equipment; and H. – Maintenance, reliability, safety factors.)*

The purpose of GARNE is to utilize new materials to modernize the radiators proposed for Nuclear Electric propulsion to be viable against other means of propulsion. While an Al/Gr radiator is constructed using novel materials for the fins, the general components would be the same as those of a standard radiator for space applications. Heat exchangers will be used to transfer heat from a controlled, high-temperature source (~1000 K) to liquid ammonia or some other heat-transfer fluid. The heat-transfer fluid will transport the heat to Al/G radiator fins, where heat can be radiated into the environment. The functional operation would consist of a radiator constructed with Al/Gr fins. Such a radiator should have an areal mass of 2-4 kg/m², in order to achieve NASA specified standards. It will be able to radiate heat effectively for an extended period of time from a source of at least 1000 K with minimal long-term damages to structural integrity or functionality. In addition to being used for radiators for nuclear-propulsion engines, having well-performing radiators will continue to be useful on a variety of spacecraft. Thermally conductive materials also have applications in electronics, such as heat-sinks, and other thermal control methods. Lightweight materials with wide-ranging applications are vital to space exploration due to large launch costs. When subjected to thermal testing, Al/G composites proved robust enough to withstand the conditions expected of a nuclear-electric propulsion system. Given that building radiators from aluminum is a straightforward process, it is reasonable to believe that a prototype can be constructed with relative ease. The Al/G composite proposed for use in radiators has a density of 1.124 g/cm³ and thermal conductivity of 280 W/mK, with the latter being 15.4% higher than that of pure aluminum. The material has a specific heat of .987 J/g/K, 9.1% higher than pure aluminum. It is proclaimed to be machinable; manufacturing radiator panels from Al/G composites is fairly straightforward. The development of this technology will require access to lab facilities. Manufacturing of materials can be done using the powder metallurgy process, which is a fairly simple way to make metallic materials and composites. A thermodynamics lab with the ability to stimulate high temperatures will be needed to test the radiator. It is difficult to know how an actual radiator made of this novel material will hold up without constructing and testing a prototype. In order to mitigate the risk of injury, simulations will be run to assess the structural and thermal capabilities of the material before any tests are run. When thermal load is applied to the radiator, it is to be done in a controlled environment. Given that the material's performance at the high temperatures necessary to be used in nuclear-electric propulsion is unknown over long timescales, close monitoring of the prototype while testing will occur to demonstrate whether or not scaling it to large radiators is viable.

SECTION III – UNIQUE OR NOVEL FEATURES OF THE INNOVATION AND THE RESULTS OR BENEFITS OF ITS APPLICATION *(Enter as appropriate: A. – Novel or unique features; B. – Advantages of innovation/software; C. – Development or new conceptual problems; D. – Test data and source of error; E. – Analysis of capabilities; and F. – For software, any re-use or re-engineering of existing code, use of shareware, or use of code owned by a non-federal entity.)*

The use of Aluminum-Graphene composites in radiators is the first proposed solution that uses graphene to solve the inefficient mass/heat expulsion problem for nuclear electric propulsion. This proposed material exhibits the best of both graphene and aluminum: having a high heat capacity and low density compared to other state-of-the-art materials. These properties outline the composite's ability to radiate heat effectively with lower mass.

SECTION IV – SPECULATION REGARDING POTENTIAL COMMERCIAL APPLICATIONS AND POINTS OF CONTACT *(Including names of companies producing or using similar products.)*

There are many potential uses of this product, including household radiators, performance vehicle companies, and commercial spaceflight companies. Some potential companies are Boeing, Aerojet Rocketdyne, SpaceX, Ferrari, and DeWitts Radiator.

10. ADDITIONAL DOCUMENTATION *(Include copies or list below any pertinent documentation which aids in the understanding or application of the innovation (e.g., articles, contractor reports, engineering specs, assembly/manufacturing drawings, parts or ingredients list, operating manuals, test data, assembly/manufacturing procedures, etc.).)*

TITLE	PAGE	DATE
Lightweight, High-Temperature Radiator for In-Space Nuclear		11/2014
Damage-Tolerant, Lightweight, High-Temperature Radiator for Nuclear Powered Spacecraft		02/2012

11. DEGREE OF TECHNOLOGY SIGNIFICANCE *(Which best expresses the degree of technological significance of this innovation?)*

☒ Modification to Existing Technology ☐ Substantial Advancement in the Art ☐ Major Breakthrough

12. STATE OF DEVELOPMENT

☒ Concept Only ☐ Design ☐ Prototype ☐ Modification ☐ Production Model ☐ Used in Current Work

13. PATENT STATUS *(Prior patent on/or related to this innovation.)*

☐ Application Filed Application No. Application Date
☐ Patent Issued Patent No. Issue Date

14. INDICATE THE DATE OR THE APPROXIMATE TIME PERIOD WHICH THIS INNOVATION WAS DEVELOPED (*i.e., conceived, constructed, tested, etc.*)
05/28/2020 - 07/09/2020

15. PREVIOUS OR CONTEMPLATED PUBLICATION OR PUBLIC DISCLOSURE INCLUDING DATES (*Provide as applicable: A. – Type of publication or disclosure, e.g., report, conference or seminar, oral presentation; B. – Disclosure by NASA or Contractor/Grantee; and C. – Title, volume no., page no., and date of publication.*)

This composite will be disclosed through a grant proposal to NASA within ASU's NPWEE class on 07/09/2020.

16. QUESTIONS FOR SOFTWARE ONLY

- (a) Using non-NASA employees to beta-test the program? ☐ YES ☐ NO If Yes, done under a beta-test agreement? ☐ YES ☐ NO
- (b) Modification of this program continued by civil servant and/or contractual agreement? ☐ YES ☐ NO
- (c) Copyright registered? ☐ YES ☐ NO ☐ UNKNOWN If Yes, then by whom?
- (d) Has the latest version been distributed outside of NASA or contractor? ☐ YES ☐ NO ☐ UNKNOWN
If Yes, date of first disclosure:
- (e) Were prior versions distributed outside of NASA or Contractor? ☐ YES ☐ NO
If Yes, supply NASA or contractor contract:
- (f) Contains or based on code not owned by U.S. Government or its contractors? ☐ YES ☐ NO ☐ UNKNOWN
If Yes, name of code and code's owner:
- Has a license for use been obtained? ☐ YES ☐ NO ☐ UNKNOWN

17. DEVELOPMENT HISTORY			
STAGE OF DEVELOPMENT	DATE (MM/YYYY)	LOCATION	IDENTIFY SUPPORTING WITNESSES (NASA in-house only)
a. First disclosure to others	06/2020	Arizona State University NPWFF Class	John Dankonich, Dann Garcia, Sheri Boonstra
b. First sketch, drawing, logic chart or code	07/2020	Arizona State University NPWFF Class	John Dankonich, Dann Garcia, Sheri Boonstra
c. First written description	07/2020	Arizona State University NPWFF Class	John Dankonich, Dann Garcia, Sheri Boonstra
d. Completion of first model of full size device (<i>invention</i>) or beta version (<i>software</i>)			
e. First successful operational test (<i>invention</i>) or alpha version (<i>software</i>)			
f. Contribution of innovators (<i>if jointly developed, provide the contribution of each innovator</i>) 1.) Jordan Levin: Principal Investigator; came up with the idea of using Aluminum-Graphene composites in nuclear electric propulsion radiators, and managed the team that explored the possibilities and scope of this idea.			

g. Indicate any past, present, or contemplated government use of the innovation

Nuclear electric propulsion was selected as the best option for a proposed Jupiter flyby mission, and implementation of GARNE could aid in development of this mission.

18. SIGNATURES OF INNOVATOR(S), WITNESS(ES), AND NASA APPROVAL

TYPED NAME AND SIGNATURE (<i>Innovator #1</i>) Jordan Levin		DATE 07/09/2020	TYPED NAME AND SIGNATURE (<i>Innovator #2</i>) Andrew Mills	DATE 07/09/2020
TYPED NAME AND SIGNATURE (<i>Innovator #3</i>) Shannon Comstock		DATE 07/09/2020	TYPED NAME AND SIGNATURE (<i>Innovator #4</i>) Candace Do	DATE 07/09/2020
TYPED NAME AND SIGNATURE (<i>Witness #1</i>) Jordan Ando		DATE 07/09/2020	TYPED NAME AND SIGNATURE (<i>Witness #2</i>) Harsha Antony	DATE 07/09/2020
NASA APPROVED	TYPED NAME	SIGNATURE		DATE

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